

YASH GOYAL

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🌐 Portfolio

WORK EXPERIENCE

Argon Robotics

June 2026 – Present

Founding Engineering Intern

USA (remote)

- Built a **real-time bimanual teleoperation stack** driving 2×6 -DOF AgileX PiPER arms from a **Meta Quest**: retargets VR controller poses through a differential-IK solver into a **100 Hz SocketCAN** control loop, guarded by per-joint slew limiting, pose-jump rejection, and hold-on-fault fallback. Isolated video and recording into separate OS processes with **shared-memory handoff** so encoding can never stall the control loop, and sequence-numbered every controller frame to surface link drops.
- Owned the **robotics data & observability pipeline** on **GCP**: concurrent real-time video feeds, **cloud RTC** to the operator dashboard plus an isolated **out-of-process recorder** with **hardware-accelerated encoding RGBD** episodes archived to **GCS buckets** and arm/operator telemetry streamed to **BigQuery** with **Prometheus + Grafana** dashboards and **~25 Slack alert rules** covering CAN faults, thermals, camera health, and operator-link drops.
- Built an end-to-end **data-to-deployment pipeline** for bimanual robot manipulation, curating **500+ teleoperation episodes** into **LeRobot** datasets on **HuggingFace**; trained and benchmarked **ACT** and **$\pi 0.5$ VLA** policies and deployed them on the same **AgileX PiPER** arms used for collection.

Emergent Labs

Mar 2026 – May 2026

Forward Deployed Engineering Intern

Bangalore

- Served as primary technical engineer for **50+ customers** running production apps, diagnosing and resolving Kubernetes (pod crashloops/scheduling), reverse-proxy, and container runtime failures across **~15 clusters**, cutting average support turnaround **~45%** and resolving **200+ infra incidents** while keeping high-traffic apps online at **99.9% uptime**.
- Built internal tooling for analytics, data, and ops teams, automating repetitive ETL and reporting workflows with [Airflow/cron jobs], and overhauling observability by instrumenting services with Prometheus metrics, Grafana dashboards, cutting mean-time-to-diagnosis and driving **~30%** productivity gains across engineering and support.

Successship Technologies

Jan 2025 – July 2025

Software Engineering Intern

Bangalore

- Improved system reliability and cut server costs **~20%** by triaging critical production issues and optimizing Docker images across **5+ services** through [base-image/multi-stage] build strategies and version upgrades.
- Led end-to-end development of a **distributor management platform** for direct-selling businesses, architecting core inventory (statistical analysis & classification), **ledger/double-entry accounting**, and **payout services** [Spring Boot, Redis, PostgreSQL]; drove the company's **first client signing** and first successful payout through the system.
- Built an AI-powered AP/AR automation system from scratch using [LLM/OCR, LangChain/OpenAI], cutting manual invoice processing by **~40%**; ran **4+ client demos** that validated usability and shaped early product direction.

PROJECTS

Axon | NestJS, React, PostgreSQL, Prisma, Redis, Gemini, Docker, GCP 🔄 🌐

- Built a **multi-tenant WhatsApp automation SaaS** letting non-technical teams design chatbot conversations on a **drag-and-drop canvas** across **8 node types**, with workspace-scoped **RBAC (4 roles)**, and **immutable published flow versions**
- Engineered a **provider-agnostic conversation engine** serving **Meta Cloud API**, **Twilio**, and a credential-free in-app **sandbox** through one code path, with **HMAC-SHA256/SHA1** webhook signature verification, **Redis-backed redelivery de-duplication**, **24-hour session windows**, and **Gemini fallback** answering off-script queries inside a configured persona.
- Deployed as a **single Docker image** (API + SPA, one port) on **GCP** behind **Caddy**, with **quota metering** before every send, **AES-256-GCM credential encryption**, and **rotating refresh tokens**.

Wanderlust | AWS EC2, Kubernetes (self-managed), Docker, NodeJS, MongoDB, Redis 🔄

- Deployed a **3-tier cloud-native application** on a **self-managed Kubernetes cluster** provisioned on AWS EC2; orchestrated workloads with Kubernetes Deployments, Services, and ConfigMaps.
- Designed the setup with a **DevOps-first approach**, focusing on infrastructure reliability & container orchestration.

SKILLS

Programming Languages: Python, Java, JavaScript

Libraries & Frameworks: ReactJS, Node.js, ExpressJS, Spring Boot

Databases: SQL, MongoDB

Cloud & DevOps: CI/CD, Cloudflare, Linux, Nginx, Docker, OTEL, ECR, ECS, Kubernetes

Core CS: Object-Oriented Programming, Data Structures & Algorithms

EDUCATION

Rishihood University, Haryana

Bachelor of Technology (CS & AI)

Aug 2023 – Present

CGPA: 8.5/10

OPEN SOURCE ( PR Dashboard)

OWASP Nest

- Enhanced the **frontend/UI** with new features for a better user experience; implemented **Django models** (with test files) for new features to ensure reliability and maintainability; shipped **SlackBot commands** to improve automation and team collaboration; and added **schema validations** while removing dead code for data integrity and a cleaner codebase.

LitmusChaos (*Organization Member*)

- Strengthened **CI quality gates** through **Docker linting** integration, enhanced local setup scripts to **streamline contributor onboarding**, and maintained repository consistency via branding and asset updates.

Palisadoes Foundation (*Organization Member*)

- Stabilized **CI/CD pipelines** with **reusable workflows** and advanced automation, improving build reliability and deployment consistency; led a **large-scale repository migration**; optimized **container images** for faster builds; and implemented production-grade **observability with OpenTelemetry** for improved monitoring and traceability.